

A Conditional Compact-Support Obstruction for Planar Heat-Conducting Compressible Flow

Abstract

We consider the planar full compressible Navier–Stokes system with positive heat conductivity and nonnegative temperature. In the strong solution class displayed in Wang’s 2022 preprint, we prove a conditional obstruction to global solutions with compactly supported initial density, positive mass, and positive physical energy, under the strict viscosity condition $\mu + \lambda > 0$. Finite spatial Lebesgue norms force the temperature and velocity to vanish in the transported exterior vacuum. A common set of admissible times then fixes the initial support ball. Compact support supplies the time regularity needed to recover the initial kinetic energy from the conservative momentum trace; the thermal trace is handled directly by the conservative equation. Conservation of total energy and the virial identity give an explicit upper bound for every interval on which the stated class is maintained. A smooth compatible family has fixed positive mass, common bounds for all the stipulated initial norms, and positive energy tending to zero. Local solvability of this family in the specified class is not established here.

Keywords Compressible Navier–Stokes equations · Heat conduction · Planar vacuum · Compact support · Physical energy

1 The equations and the conditional obstruction

We study the Cauchy problem for the heat-conducting compressible Navier–Stokes system on $\mathbb{R}^2$:

$$\rho_t + \operatorname{div}(\rho u) = 0, \quad (1.1)$$

$$(\rho u)_t + \operatorname{div}(\rho u \otimes u) - \operatorname{div} S + \nabla P = 0, \quad (1.2)$$

$$c_v [(\rho \theta)_t + \operatorname{div}(\rho \theta u)] = Q(u) - P \operatorname{div} u + \kappa \Delta \theta. \quad (1.3)$$

Here ρ , $u = (u_1, u_2)$, and θ denote the density, velocity, and absolute temperature. The pressure and viscous stress are

$$\begin{aligned} P &= R\rho\theta, & D(u) &= \frac{1}{2}(\nabla u + \nabla u^\top), \\ S &= 2\mu D(u) + \lambda(\operatorname{div} u)I, & Q(u) &= S : \nabla u = 2\mu|D(u)|^2 + \lambda(\operatorname{div} u)^2. \end{aligned} \quad (1.4)$$

We assume throughout that

$$\mu > 0, \quad b := \mu + \lambda > 0, \quad R, c_v, \kappa > 0. \quad (1.5)$$

The source condition [1, (1.2)] permits $b \geq 0$; our conclusion is restricted to $b > 0$. In particular, $\operatorname{div} S = \mu \Delta u + b \nabla \operatorname{div} u$. The letter R always denotes the gas constant; support balls will have radius L .

The argument concerns a precise solution class. In [1, Theorem 1.1], Wang states local existence for this system with vacuum and gives a complete list of weighted

and unweighted regularity properties. We use that displayed list as a hypothesis on a putative solution and do not invoke the local existence assertion to construct one. Thus the conclusion below excludes global solutions in the stated class. Finite-time breakdown of a particular locally constructed solution additionally requires a separate local construction.

The exterior thermal equation is decisive. In a spacetime vacuum region it becomes $-\kappa\Delta\theta = Q(u)$. In two dimensions, nonnegativity and finite spatial Lebesgue integrability force such a superharmonic temperature to vanish on a connected exterior domain. Under (1.5), the resulting equality $Q(u) = 0$ makes the velocity a rigid motion there, and its finite Lebesgue norm eliminates that motion. We first carry out this argument on the transported exterior of the initial support ball. After proving that the exterior flow is stationary, we use the resulting fixed support bound in the energy and virial identities.

All function spaces and spatial integrals without a specified domain are over $\mathbb{R}^2$. On a fixed finite time interval, constants in intermediate estimates may depend on that interval, the physical parameters, the exponents, the support radius, and the finite solution norms explicitly used. The constant in the final lifespan bound is given exactly.

Fix $a > 1$, $q > 2$, and $\eta_0 > 0$, and set

$$\begin{aligned}\bar{x} &= (e + |x|^2)^{1/2} \log^{1+\eta_0}(e + |x|^2), \\ \alpha &= \min\{a/2, 1\}, \quad p_u = \frac{4}{\alpha}, \quad p_\theta = \frac{6}{2\alpha - 1}.\end{aligned}\tag{1.6}$$

Thus $1/2 < \alpha \leq 1$, $4 \leq p_u < 8$, and $6 \leq p_\theta < \infty$. The exponents p_u, p_θ are denoted by q_2, q_3 in [1, (1.9)]. We also fix a finite $q_1 > 2$ as in that display.

The initial assumptions are the displayed data conditions [1, (1.6)–(1.7)]:

$$\begin{aligned}\rho_0, \theta_0 &\geq 0, \quad \bar{x}^a \rho_0 \in L^1 \cap H^1 \cap W^{1,q}, \\ \sqrt{\rho_0} u_0, \sqrt{\rho_0} \theta_0 &\in L^2, \quad \nabla u_0 \in H^1, \quad \nabla \theta_0 \in L^2,\end{aligned}\tag{1.7}$$

and

$$-\mu\Delta u_0 - b\nabla \operatorname{div} u_0 + R\nabla(\rho_0 \theta_0) = \sqrt{\rho_0} g, \quad g \in L^2.\tag{1.8}$$

In particular, $\|\sqrt{\rho_0} \theta_0\|_2^2 = \int \rho_0 \theta_0^2 dx$. No unweighted L^2 assumption on u_0 or θ_0 , finite- L^{p_u} assumption on u_0 , or thermal compatibility condition is included in these displayed data conditions.

Definition 1.1 A solution in the specified class on $[0, T_*)$, where $T_* > 0$, consists of ρ, u, θ with $\rho, \theta \geq 0$, for which all derivatives in (1.1)–(1.3) are regular distributions and the equations hold almost everywhere. For every $0 < T < T_*$ it satisfies the

complete regularity list

$$\begin{aligned}
\rho &\in C([0, T]; L^1 \cap H^1 \cap W^{1,q}), \\
\bar{x}^\alpha \rho &\in L^\infty(0, T; L^1 \cap H^1 \cap W^{1,q}), \\
\nabla u &\in L^\infty(0, T; H^1) \cap L^2(0, T; W^{1,q}), \\
u &\in L^\infty(0, T; L^{p_u}), \\
\sqrt{\rho} u, \sqrt{\rho} \theta, \sqrt{\rho} u_t, \nabla \theta, \sqrt{t} \sqrt{\rho} \theta_t, \\
&\quad \sqrt{t} \nabla^2 \theta, \bar{x}^{\alpha/2} \nabla u \in L^\infty(0, T; L^2), \\
\nabla u_t, \sqrt{\rho} \theta_t, \nabla^2 \theta, \sqrt{t} \nabla \theta_t &\in L^2(0, T; L^2), \\
\theta &\in L^2(0, T; L^{p_\theta}), \quad \sqrt{t} \nabla^2 \theta \in L^2(0, T; L^{q_1}).
\end{aligned} \tag{1.9}$$

Its initial conditions are imposed on conservative variables:

$$\begin{aligned}
\rho(0) &= \rho_0, \quad \rho u(t) \longrightarrow \rho_0 u_0, \quad \rho \theta(t) \longrightarrow \rho_0 \theta_0 \\
&\quad \text{in } \mathcal{D}'(\mathbb{R}^2) \quad (t \downarrow 0).
\end{aligned} \tag{1.10}$$

The density trace is also strong in the spaces specified in (1.9).

The list (1.9) reproduces [1, (1.8)]; the interpretation of strong derivatives is [1, Definition 1.1]. The conservative initial variables are those of [1, (1.3)]. The far-field condition [1, (1.4)] is qualified by *in some weak sense*. We do not assign an additional topology to that phrase. Our proof uses the explicit finite norms of u and θ in (1.9), imposed on the functions themselves. Replacing them by conditions on representatives modulo constants would change the hypothesis.

For clarity, the following portion of the full class suffices for the proof:

$$\begin{aligned}
\rho &\in C([0, T]; L^1 \cap H^1 \cap W^{1,q}), \\
\nabla u &\in L^\infty(0, T; H^1) \cap L^2(0, T; W^{1,q}), \\
u &\in L^\infty(0, T; L^{p_u}), \\
\sqrt{\rho} u, \sqrt{\rho} \theta, \nabla \theta &\in L^\infty(0, T; L^2), \\
\nabla u_t, \nabla^2 \theta &\in L^2(0, T; L^2), \quad \theta \in L^2(0, T; L^{p_\theta}).
\end{aligned} \tag{1.11}$$

This identifies the analytic inputs; Definition 1.1 retains the full original list. The unweighted spacetime Hessian bound in (1.11) is an explicit source assumption, not a consequence asserted from its time-weighted counterpart.

Suppose now that, for some $L > 0$,

$$\begin{aligned}
\text{supp } \rho_0 &\subset \overline{B_L}, \quad M := \int \rho_0 dx > 0, \\
E_0 &:= \int (\tfrac{1}{2} \rho_0 |u_0|^2 + c_v \rho_0 \theta_0) dx > 0.
\end{aligned} \tag{1.12}$$

Positive mass is explicit here; no mass normalization is used. The energy is finite by (1.7), since $\int \rho_0 \theta_0 dx \leq M^{1/2} \|\sqrt{\rho_0} \theta_0\|_2$. Define

$$I_0 = \int \rho_0 |x|^2 dx, \quad J_0 = \int \rho_0 u_0 \cdot x dx, \quad \beta = \min\{1, R/c_v\} > 0. \tag{1.13}$$

Both moments are finite: $I_0 \leq ML^2$ and $|J_0| \leq I_0^{1/2} \|\sqrt{\rho_0} u_0\|_2$. In fact,

$$I_0 < ML^2, \quad (1.14)$$

because ρ_0 is an integrable function of positive mass and the circle $\{|x| = L\}$ has two-dimensional Lebesgue measure zero.

Theorem 1.1 *Assume (1.5), (1.7), (1.8), and (1.12). If a solution in the sense of Definition 1.1 exists on $[0, T_*)$, then for every $0 \leq t < T_*$,*

$$I_0 + 2J_0 t + 2\beta E_0 t^2 \leq \int \rho(t, x) |x|^2 dx \leq ML^2. \quad (1.15)$$

Consequently,

$$T_* \leq \frac{-J_0 + \sqrt{J_0^2 + 2\beta E_0 (ML^2 - I_0)}}{2\beta E_0} < \infty. \quad (1.16)$$

In particular, no global solution satisfying the stated class on every finite interval exists for these data.

The theorem requires neither nonzero total momentum nor a positive density or temperature lower bound. Its second moment follows from compact support. If a local solution has been constructed independently, (1.16) limits any extension maintaining this class. Without that construction, the conclusion remains the conditional absence of a global solution.

2 Exterior vacuum and fixed support

We first prove the planar rigidity fact used for the temperature.

Lemma 2.1 *Let $\Omega \subset \mathbb{R}^2$ be connected and open, and suppose that $\{|x| > L_1\} \subset \Omega$ for some $L_1 > 0$. If*

$$v \in W_{\text{loc}}^{2,2}(\Omega) \cap L^p(\Omega), \quad v \geq 0, \quad -\Delta v \geq 0$$

in distributions, where $1 \leq p < \infty$, then $v = 0$ in Ω .

Proof Local Sobolev regularity gives a continuous representative of v . For $s > \log L_1$, let

$$h(s) = \frac{1}{2\pi} \int_0^{2\pi} v(e^s \cos \varphi, e^s \sin \varphi) d\varphi.$$

The polar change of variables on compact annuli and angular integration show that $h \in H_{\text{loc}}^2$ and

$$h''(s) = \frac{e^{2s}}{2\pi} \int_0^{2\pi} \Delta v(e^s \cos \varphi, e^s \sin \varphi) d\varphi \leq 0 \quad (2.1)$$

almost everywhere. Thus h is nonnegative and concave on a half-line. It is nondecreasing: a negative secant slope, together with concavity, would force h to become negative at sufficiently large s . Jensen's inequality gives

$$2\pi \int_{\log L_1}^{\infty} h(s)^p e^{2s} ds \leq \int_{|x| > L_1} |v(x)|^p dx < \infty. \quad (2.2)$$

If $h(s_0) > 0$, monotonicity would make the left side infinite. Hence $h = 0$, and nonnegativity and continuity imply $v = 0$ on $\{|x| > L_1\}$.

To propagate this equality, the local super-mean inequality gives

$$\frac{1}{2\pi} \int_0^{2\pi} v(z + r(\cos \varphi, \sin \varphi)) d\varphi \leq v(z), \quad \overline{B_r(z)} \subset \Omega. \quad (2.3)$$

For the stated regularity, convolve v inside a slightly larger ball, apply the smooth inequality, and pass to the limit locally uniformly. If $v(z) = 0$, nonnegativity forces the average in (2.3) to vanish for every sufficiently small r . Continuity therefore gives $v = 0$ in a neighborhood of z . The zero set is relatively open and closed in Ω and is nonempty. Connectedness proves the assertion. $\square$

Proposition 2.2 *Let a solution satisfy Definition 1.1 on $[0, T_*)$, and suppose that $\text{supp } \rho_0 \subset \overline{B_L}$. Then*

$$\begin{aligned} \text{supp } \rho(t) &\subset \overline{B_L} \quad (0 \leq t < T_*), \\ u(t) = \theta(t) &= 0 \quad \text{on } \mathbb{R}^2 \setminus \overline{B_L} \end{aligned} \quad (2.4)$$

for almost every $t \in (0, T_*)$. The density assertion holds for its continuous spatial representative.

Proof Fix $0 < T < T_*$. Since $q > 2$, Sobolev embedding applied to ∇u gives $\|\nabla u(t)\|_\infty \leq C_q \|\nabla u(t)\|_{W^{1,q}}$ for almost every t . Averaging the velocity over a unit ball and using its Lipschitz bound, we obtain

$$\begin{aligned} |u(t, x)| &\leq \frac{1}{|B_1|} \int_{B_1(x)} |u(t, y)| dy + C \|\nabla u(t)\|_\infty \\ &\leq C_{pu} \|u(t)\|_{p_u} + C \|\nabla u(t)\|_\infty. \end{aligned} \quad (2.5)$$

The source norms and Cauchy–Schwarz in time imply

$$A_T := \int_0^T \|u(t)\|_\infty dt < \infty, \quad B_T := \int_0^T \|\nabla u(t)\|_\infty dt < \infty. \quad (2.6)$$

The spatial representative is C^1 for almost every time, since $\nabla u \in W^{1,q}$. Values on the remaining null set of times do not affect the integral flow

$$X(t, y) = y + \int_0^t u(s, X(s, y)) ds. \quad (2.7)$$

Existence and forward and backward uniqueness follow from (2.6). Gronwall's inequality gives

$$\begin{aligned} |X(t, y) - y| &\leq A_T, \\ e^{-B_T} |y - z| &\leq |X(t, y) - X(t, z)| \leq e^{B_T} |y - z|. \end{aligned} \quad (2.8)$$

Thus $X(t, \cdot)$ is a bi-Lipschitz homeomorphism. The flow and its inverse are jointly continuous in space and time, and

$$\mathcal{J}(t, y) = \det \nabla_y X(t, y) = \exp \left(\int_0^t \text{div } u(s, X(s, y)) ds \right) > 0. \quad (2.9)$$

The variational equation for spatial derivatives is justified by the spatial C^1 representative and the integrable derivative bound; it also follows by spatial smoothing and Lipschitz stability of the flow. In particular, $\mathcal{J}$ has positive upper and lower bounds depending only on B_T .

We justify the density characteristic identity rather than assume a pointwise time chain rule. For a spatial mollifier ζ_δ put $\rho_\delta = \zeta_\delta * \rho$. The continuity equation yields

$$(\rho_\delta)_t + u \cdot \nabla \rho_\delta + (\operatorname{div} u) \rho_\delta = r_\delta. \quad (2.10)$$

Its transport commutator satisfies

$$\|u \cdot \nabla \rho_\delta - (u \cdot \nabla \rho)_\delta\|_q \leq C\delta \|\nabla u\|_\infty \|\nabla \rho\|_q. \quad (2.11)$$

The other commutator, $(\operatorname{div} u) \rho_\delta - ((\operatorname{div} u) \rho)_\delta$, tends to zero in L^q for almost every time and is bounded by $2\|\operatorname{div} u\|_\infty \|\rho\|_q$. Hence $r_\delta \rightarrow 0$ in $L^1(0, T; L^q)$. Integrate (2.10) along (2.7). The Jacobian bounds allow passage to the limit in L_y^q , giving

$$\rho(t, X(t, y)) \mathcal{J}(t, y) = \rho_0(y). \quad (2.12)$$

Initially this holds almost everywhere in y . Spatial continuity of ρ , the flow, and (2.9) extends it to every y . Continuity of ρ in time in $W^{1,q}$ extends it to every $t \in [0, T]$. No division by density is involved.

Put $\Omega_0 = \mathbb{R}^2 \setminus \overline{B_L}$ and $\Omega_t = X(t, \Omega_0)$. Equation (2.12) gives $\rho(t) = 0$ on Ω_t for every t . Each Ω_t is connected and open, with compact complement $X(t, \overline{B_L})$, and

$$\{|x| > L + A_T + 1\} \subset \Omega_t, \quad 0 \leq t \leq T. \quad (2.13)$$

At this stage the support is only transported. The spacetime set

$$\mathcal{V} = \{(t, x) : 0 < t < T, X(t, \cdot)^{-1}(x) \in \Omega_0\}$$

is open by joint continuity of the inverse flow. On $\mathcal{V}$, $\rho = 0$, so $\rho\theta$, $\rho\theta u$, and P vanish as distributions. Restriction of the conservative thermal equation gives

$$-\kappa \Delta \theta = Q(u) \quad \text{in } \mathcal{D}'(\mathcal{V}). \quad (2.14)$$

This uses an open spacetime vacuum region; it does not set a time derivative to zero merely from vacuum on a single time slice.

To slice (2.14), cover $\mathcal{V}$ by countably many spacetime cylinders with closures contained in $\mathcal{V}$. For a countable dense collection of spatial tests in each cylinder, Fubini's theorem removes a single null set of times. Taking the countable union of these exceptional sets, together with the null sets for the source norms and signs, gives one full-measure set $G \subset (0, T)$ such that, for each $t \in G$,

$$\begin{aligned} \theta(t) &\in W_{\text{loc}}^{2,2}(\mathbb{R}^2) \cap L^{p_\theta}, & \theta(t) &\geq 0, \\ u(t) &\in L^{p_u}, & \nabla u(t) &\in W^{1,q}, \\ -\kappa \Delta \theta(t) &= Q(u(t)) \quad \text{in } \mathcal{D}'(\Omega_t). \end{aligned} \quad (2.15)$$

Local L^2 integrability of $\theta(t)$ follows from $p_\theta \geq 2$; the gradient and unweighted Hessian bounds provide the other local derivatives. Thus $\theta(t)$ is continuous. Crucially, G has been chosen independently of all trajectories and spatial points.

Write $d = \operatorname{div} u$ and $D_0 = D(u) - \frac{1}{2}dI$. In two dimensions,

$$Q(u) = 2\mu|D_0|^2 + bd^2 \geq 0. \quad (2.16)$$

For every $t \in G$, Lemma 2.1 applies on Ω_t with $p = p_\theta$. It gives $\theta(t) = 0$ there. Then (2.14) gives $Q(u(t)) = 0$ there, and strict $\mu > 0, b > 0$ imply $D(u(t)) = 0$. The distributional identity

$$\partial_j \partial_k u_i = \partial_j D_{ik}(u) + \partial_k D_{ij}(u) - \partial_i D_{jk}(u) \quad (2.17)$$

shows that all second derivatives of $u(t)$ vanish on Ω_t . Connectedness gives $u(t, x) = A(t)x + c(t)$, where $A(t)$ is a constant skew-symmetric matrix. A nonzero constant or planar rotation has infinite L^{p_u} norm on a full exterior region. Hence

$$u(t, x) = \theta(t, x) = 0 \quad (x \in \Omega_t, t \in G). \quad (2.18)$$

Spatial continuity makes these equalities pointwise.

Fix any $y \in \Omega_0$. Since $X(t, y) \in \Omega_t$, (2.18) gives $u(t, X(t, y)) = 0$ on the same set G for every such y . Absolute continuity of the trajectories yields

$$X(t, y) = y \quad (0 \leq t \leq T, y \in \Omega_0).$$

Continuity fixes ∂B_L as well. Bijectivity therefore implies $X(t, \overline{B_L}) = \overline{B_L}$ and $\Omega_t = \Omega_0$. Equations (2.12) and (2.18) now prove (2.4). The argument applies to every $T < T_*$. $\square$

The fixed initial ball is essential: a radius growing with $\int_0^t \|u\|_\infty$ would not give the uniform moment bound in (1.15). Strict viscosity is also essential to this proof. At $b = 0$, (2.16) does not control $\operatorname{div} u$, so $Q(u) = 0$ does not force $D(u) = 0$. No endpoint assertion is made here.

3 Time traces and conservation of energy

We now justify the total-energy constant, including its value at $t = 0$. Conservative momentum convergence alone does not imply convergence of the kinetic energy. The additional time regularity required for that step follows from Proposition 2.2.

Proposition 3.1 *Under the hypotheses of Theorem 1.1, set*

$$K(t) = \frac{1}{2} \int \rho |u|^2 dx, \quad U(t) = c_v \int \rho \theta dx.$$

For each $T < T_$ these quantities have absolutely continuous representatives on $[0, T]$, with*

$$\begin{aligned} K(0) &= \frac{1}{2} \int \rho_0 |u_0|^2 dx, & U(0) &= c_v \int \rho_0 \theta_0 dx, \\ K'(t) &= - \int Q(u) dx + \int P \operatorname{div} u dx, \\ U'(t) &= \int Q(u) dx - \int P \operatorname{div} u dx \end{aligned} \quad (3.1)$$

for almost every t . In particular,

$$K(t) + U(t) = E_0, \quad K(t), U(t) \geq 0, \quad 0 \leq t < T_*. \quad (3.2)$$

Moreover, $\int \rho(t) dx = M$ for all $t < T_*$.

Proof Fix $T < T_*$. The fixed support, the existing whole-space weak derivatives, and the finite spatial Lebesgue norms permit the Poincaré inequality on a slightly larger ball. Thus

$$\begin{aligned} u &\in L^\infty(0, T; H^2), \\ \theta &\in L^\infty(0, T; H^1) \cap L^2(0, T; H^2). \end{aligned} \quad (3.3)$$

Indeed, $\|u\|_2 \leq C_L \|\nabla u\|_2$ and $\|\theta\|_2 \leq C_L \|\nabla \theta\|_2$; the remaining derivatives are already in (1.11). These are whole-space Sobolev functions, so the argument introduces no boundary conditions on an unknown vacuum interface.

The distribution u_t also vanishes outside B_L in spacetime. We claim that

$$\|u_t\|_{L^2(0, T; L^2)} \leq C_L \|\nabla u_t\|_{L^2(0, T; L^2)}, \quad u_t \in L^2(0, T; H^1). \quad (3.4)$$

To prove this for the distributional time derivative, convolve first in time on a compact subinterval of $(0, T)$ and then in space with a mollifier of radius $\delta < 1$. The resulting smooth function $w_{\varepsilon, \delta}$ is supported in $B_{L+\delta}$ spatially. The Poincaré inequality on B_{L+1} gives

$$\|w_{\varepsilon, \delta}(t)\|_2 \leq C_L \|\nabla w_{\varepsilon, \delta}(t)\|_2.$$

Integration on the compact time interval and the L^2 contraction of convolution bound this by the given spacetime norm of ∇u_t . Weak compactness and distributional convergence identify the limit with u_t . Exhausting $(0, T)$ by compact subintervals proves (3.4). The constant is independent of the smoothing parameters and the compact time subinterval.

Consequently, $u \in H^1(0, T; H^1) \cap L^\infty(0, T; H^2)$, and its continuous H^1 representative has a limit

$$u(t) \longrightarrow u_* \quad \text{in } H^1 \quad (t \downarrow 0). \quad (3.5)$$

Since $\rho(t) \rightarrow \rho_0$ in $W^{1, q}$ and hence in L^∞ , we have $\rho(t)u(t) \rightarrow \rho_0 u_*$ in L^2 . The conservative momentum trace identifies

$$\rho_0 u_* = \rho_0 u_0 \quad \text{almost everywhere.} \quad (3.6)$$

This imposes no identification on the initial vacuum set. For the kinetic energy,

$$\begin{aligned} &\left| \int \rho(t) |u(t)|^2 dx - \int \rho_0 |u_*|^2 dx \right| \\ &\leq \|\rho(t) - \rho_0\|_\infty \|u(t)\|_2^2 \\ &\quad + \|\rho_0\|_\infty \|u(t) - u_*\|_2 (\|u(t)\|_2 + \|u_*\|_2) \longrightarrow 0. \end{aligned} \quad (3.7)$$

Equation (3.6) then gives the asserted value of $K(0)$.

We derive the balances with integrable right sides. Continuity and the preceding regularity imply

$$\rho_t = -u \cdot \nabla \rho - \rho \operatorname{div} u \in L^2(0, T; L^q). \quad (3.8)$$

Here $u \in L^\infty(0, T; L^\infty)$ follows from (3.3); the other factors use $\rho, \nabla \rho \in L^\infty(0, T; L^q)$ and $\operatorname{div} u \in L^2(0, T; L^\infty)$. The time product rule for $\int \rho |u|^2$ is legitimate: for the ρ_t term use $|u|^2 \in L^\infty(0, T; L^{q/(q-1)})$, and for the u_t term use $\rho \in L^\infty$ and $u, u_t \in L^2$ in space. These bounds also justify passage from space and time mollifications to the unmollified products.

Using continuity to write momentum as $\rho(u_t + u \cdot \nabla u) = \operatorname{div} S - \nabla P$, we test by u , with a fixed spatial cutoff equal to one near $\overline{B_L}$. The convective term obeys

$$\int \rho(u \cdot \nabla u) \cdot u dx = -\frac{1}{2} \int \operatorname{div}(\rho u) |u|^2 dx = \frac{1}{2} \int \rho_t |u|^2 dx. \quad (3.9)$$

Weak integration by parts gives

$$\int \operatorname{div} S \cdot u dx = - \int Q(u) dx, \quad - \int \nabla P \cdot u dx = \int P \operatorname{div} u dx.$$

These pairings are valid with $S, P \in L^2$ and $u \in H^1$. In fact,

$$\|P(t)\|_2 \leq R \|\rho(t)\|_\infty^{1/2} \|\sqrt{\rho(t)} \theta(t)\|_2. \quad (3.10)$$

Thus

$$\frac{d}{dt} \frac{1}{2} \int \rho |u|^2 dx = - \int Q(u) dx + \int P \operatorname{div} u dx. \quad (3.11)$$

The bounds

$$\begin{aligned} \int |Q(u)| dx &\leq C(\mu, \lambda) \|\nabla u\|_2^2, \\ \left| \int P \operatorname{div} u dx \right| &\leq \|P\|_2 \|\operatorname{div} u\|_2 \end{aligned} \quad (3.12)$$

show that the right side is integrable on $[0, T]$. Together with (3.7), this proves the asserted absolute continuity of K .

For internal energy we keep the conservative thermal equation. Choose $\chi \in C_c^\infty(\mathbb{R}^2)$ equal to one on a neighborhood of $\overline{B_L}$. Spatial pairing with χ gives

$$\frac{d}{dt} \left(c_v \int \rho \theta \chi dx \right) = \int Q(u) dx - \int P \operatorname{div} u dx. \quad (3.13)$$

Indeed, $\rho \theta u \cdot \nabla \chi = 0$. The function θ and its distributional spatial derivatives are supported in $\overline{B_L}$ for almost every time, so

$$\langle \Delta \theta, \chi \rangle = \langle \theta, \Delta \chi \rangle = 0.$$

The functions $Q(u)$ and $P \operatorname{div} u$ are supported in that ball too. Thus no flux term at infinity or at a moving vacuum boundary has been dropped.

The left side of (3.13) is $U'(t)$, and U is essentially bounded because $|\int \rho \theta dx| \leq \|\rho\|_1^{1/2} \|\sqrt{\rho} \theta\|_2$. The right side lies in $L^1(0, T)$ by (3.12), giving an absolutely continuous extension to $[0, T]$. Pairing the conservative thermal trace (1.10) with the same χ yields

$$U(0) = c_v \int \rho_0 \theta_0 dx.$$

No unweighted strong trace of θ is required. Adding the kinetic and internal balances proves (3.2), including its constant at zero. The signs follow from $\rho, \theta \geq 0$ and extend to the continuous energy representatives. Finally, testing (1.1) with χ gives conservation of mass, whose value is M by the density trace. $\square$

The equality in (3.2), with its correctly identified initial value, is needed below. An upper energy inequality alone would not give the positive lower bound for the virial acceleration. The conservative thermal test avoids any unproved temperature chain rule in vacuum.

4 The virial bound and small positive-energy data

Proof of Theorem 1.1 Fix $0 < T < T_*$. Let χ be the cutoff in Proposition 3.1 and define

$$I(t) = \int \rho(t, x) |x|^2 dx, \quad J(t) = \int \rho(t, x) u(t, x) \cdot x dx.$$

By fixed support, these are the pairings with $\chi|x|^2$ and χx . They are finite; in particular, $|J(t)| \leq LM^{1/2} \|\sqrt{\rho(t)} u(t)\|_2$. Testing continuity with $\chi|x|^2$ gives

$$I'(t) = 2J(t). \quad (4.1)$$

Testing conservative momentum with χx gives

$$J'(t) = \int \rho |u|^2 dx - \int \text{tr} S dx + 2 \int P dx. \quad (4.2)$$

Terms involving derivatives of χ are supported where $\rho = u = \theta = 0$ and their weak spatial derivatives vanish. In two dimensions, $\text{tr} S = 2(\mu + \lambda) \text{div} u = 2b \text{div} u$. Fixed support and $\nabla u \in L^2$ imply $\text{div} u \in L^1$, and

$$\int \text{div} u dx = - \int u \cdot \nabla \chi dx = 0.$$

Consequently,

$$J'(t) = 2K(t) + 2 \frac{R}{c_v} U(t). \quad (4.3)$$

The right side is integrable, so J has an absolutely continuous representative. The conservative momentum trace tested with χx gives $J(0) = J_0$; the density trace gives $I(0) = I_0$. Equation (4.1) identifies the absolutely continuous derivative of I . Nonnegativity and conservation of energy give

$$I''(t) = 4K(t) + 4 \frac{R}{c_v} U(t) \geq 4\beta (K(t) + U(t)) = 4\beta E_0 \quad (4.4)$$

almost everywhere. Integrating twice,

$$I(t) \geq I_0 + 2J_0 t + 2\beta E_0 t^2, \quad 0 \leq t \leq T. \quad (4.5)$$

On the other hand, fixed support and mass conservation yield

$$0 \leq I(t) \leq L^2 \int \rho(t) dx = ML^2. \quad (4.6)$$

This proves (1.15). Since $E_0 > 0$ and $ML^2 - I_0 > 0$, the inequality

$$2\beta E_0 t^2 + 2J_0 t - (ML^2 - I_0) \leq 0$$

has exactly one positive root, the right side of (1.16). Every $t < T_*$ is bounded by it, since $T < T_*$ was arbitrary. The theorem follows. $\square$

We finish with smooth initial data whose common norm bounds can be fixed before the energy parameter is chosen.

Corollary 4.1 *Fix the parameters in (1.5), $a > 1$, $q > 2$, $\eta_0 > 0$, and a mass $M > 0$. There is a family $(\rho_0^\varepsilon, u_0^\varepsilon, \theta_0^\varepsilon)$, $0 < \varepsilon \leq 1$, of smooth compactly supported data satisfying (1.7)–(1.8) such that*

$$\begin{aligned} \int \rho_0^\varepsilon dx &= M, & \text{supp } \rho_0^\varepsilon &\subset \overline{B_1}, \\ E_0^\varepsilon &> 0, & E_0^\varepsilon &\longrightarrow 0. \end{aligned} \quad (4.7)$$

The density is independent of ε . Every norm displayed in (1.7), as well as $\|g^\varepsilon\|_2$, has a common finite upper bound. No member of the family admits a global solution in Definition 1.1.

Proof Take the nonnegative smooth function

$$f(x) = \begin{cases} \exp(-1/(1-|x|^2)), & |x| < 1, \\ 0, & |x| \geq 1, \end{cases} \quad c = \frac{M}{\int f^2 dx}, \quad r = cf^2.$$

Then $r, \sqrt{r} \in C_c^\infty(\mathbb{R}^2)$, $r > 0$ on B_1 , and $\int r dx = M$. Set

$$\rho_0^\varepsilon = r, \quad u_0^\varepsilon = 0, \quad \theta_0^\varepsilon = \varepsilon r, \quad 0 < \varepsilon \leq 1. \quad (4.8)$$

The weighted density norm $\|\bar{x}^\alpha r\|_{L^1 \cap H^1 \cap W^{1,q}}$ is fixed and finite. The velocity norms vanish, and

$$\begin{aligned} \|\sqrt{r} \theta_0^\varepsilon\|_2 &= \varepsilon \left(\int r^3 dx \right)^{1/2}, \\ \|\nabla \theta_0^\varepsilon\|_2 &= \varepsilon \|\nabla r\|_2. \end{aligned} \quad (4.9)$$

The momentum compatibility condition holds exactly:

$$\begin{aligned} -\mu \Delta u_0^\varepsilon - b \nabla \text{div } u_0^\varepsilon + R \nabla (\rho_0^\varepsilon \theta_0^\varepsilon) \\ = R \varepsilon \nabla (r^2) = 2R \varepsilon r \nabla r = \sqrt{r} g^\varepsilon, \\ g^\varepsilon = 2R \varepsilon \sqrt{r} \nabla r. \end{aligned} \quad (4.10)$$

Since $\sqrt{r} \in C_c^\infty$, g^ε is smooth and compactly supported, with

$$\|g^\varepsilon\|_2 = 2R \varepsilon \left(\int r |\nabla r|^2 dx \right)^{1/2}. \quad (4.11)$$

All required initial bounds are therefore uniform for $0 < \varepsilon \leq 1$. No thermal compatibility condition is imposed.

Finally,

$$E_0^\varepsilon = c_v \varepsilon \int r^2 dx > 0, \quad J_0^\varepsilon = 0, \quad I_0^\varepsilon = \int r|x|^2 dx < M. \quad (4.12)$$

Theorem 1.1 applies to each $\varepsilon > 0$. If a solution in the specified class exists on $[0, T_*^\varepsilon)$, its interval satisfies

$$T_*^\varepsilon \leq \left(\frac{M - \int r|x|^2 dx}{2\beta c_v \varepsilon \int r^2 dx} \right)^{1/2}. \quad (4.13)$$

This is finite for every $\varepsilon > 0$. □

The family has fixed positive mass, fixed density support, and zero total momentum. The initial velocity and temperature are actual compactly supported smooth functions, so their integrability has no constant-representative ambiguity. A positive energy threshold based only on the fixed parameters and these common initial upper bounds therefore cannot guarantee a global solution in the stated class for every member of the family. The upper bound (4.13) tends to infinity as $\varepsilon \downarrow 0$; the conclusion concerns each strictly positive energy.

The construction verifies the displayed initial assumptions and compatibility condition, but does not independently establish local solvability in (1.9). It gives no example of finite-time breakdown without that additional result. The theorem also retains nonnegative temperature, finite unweighted velocity and temperature norms, and strict $\mu + \lambda > 0$. Different solution classes and the viscosity endpoint require separate arguments.

References

1. Wang, X.: On Local Existence and Blowup Criterion of Strong Solutions to Cauchy Problem of 2D Full Compressible Navier-Stokes System with Vacuum. arXiv:2212.13343v1 [math.AP], 27 December 2022.